

Track B = the **next-DAY** volatility forecast RV_{d+1} (the deferred Layer-3 battery). Target is next trading day realized vol; **HAR-RV is the benchmark to beat**; regime = the continuum covariance dial (LE / AI). QLIKE on variance (=vol²). Purged walk-forward by year, embargo 1 day.

987
Trading days

964
Valid rows

2023,2024,2025
Test years

Does the covariance-regime dial beat HAR at the next-DAY horizon?

Sigma	metric	R2 HAR	R2 HAR+LOB	R2 HAR+LOB+regime	dR2 (regime over HAR+LOB)	DM vs HAR p (full)	MCS90	perm survives	verdict
30s	LE	0.8298	0.8335	0.8374	0.0039	0.9268	-	YES	marginal +
30s	AI	0.8298	0.8335	0.8197	-0.0138	0.0000	-	no	does NOT beat HAR
45s	LE	0.8298	0.8335	0.8375	0.0040	0.3484	-	YES	marginal +
45s	AI	0.8298	0.8335	0.8191	-0.0144	0.0000	-	no	does NOT beat HAR
60s	LE	0.8298	0.8335	0.8342	0.0006	0.3566	-	YES	marginal +
60s	AI	0.8298	0.8335	0.8211	-0.0124	0.0000	-	no	does NOT beat HAR

HONEST READ: at the next-DAY horizon daily RV is highly persistent, so HAR already captures ~0.83 OOS R2. The dial adds at most a few tenths of a percent under LE (not DM-significant, eliminated from the MCS) and **hurts** under AI. The continuum-regime edge that shows up intraday (Track A) does NOT survive to the next-day horizon here.